

Лабораторная работа 7

Адресация IPv4 и IPv6. Настройка DHCP

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Цель работы

Изучение механизмов динамической конфигурации параметров IPv4- и IPv6-сетей с использованием DHCPv4 и DHCPv6, а также взаимодействия DHCP с SLAAC и ICMPv6 в среде моделирования GNS3.

Задачи

1. Развертывание стенда в GNS3 с маршрутизатором VyOS.
2. Настройка DHCPv4-сервера для автоматической выдачи IPv4-адресов.
3. Проверка получения адреса клиентом VPCS.
4. Настройка stateless-DHCPv6 с SLAAC.
5. Настройка stateful-DHCPv6 с полной выдачей адреса.
6. Анализ трафика ICMPv6 и DHCPv6.

Стенд

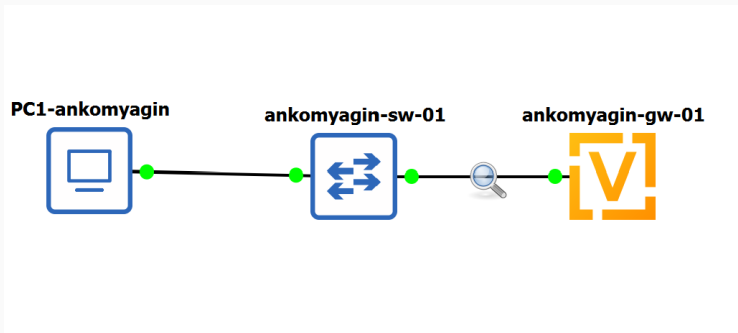


Рис. 1: Топология: маршрутизатор VyOS с тремя коммутаторами и хостами

- Маршрутизатор VyOS ankomyagin-gw-01;
- Коммутаторы: ankomyagin-sw-01, ankomyagin-sw-02, ankomyagin-sw-03;
- Хосты:
 - PC1 (VPCS) — DHCPv4 на eth0;
 - PC2 (Linux) — SLAAC + stateless-DHCPv6 на eth0;
 - PC3 (Linux) — stateful-DHCPv6 на eth0.

Интерфейс	IPv4	IPv6
eth0	10.0.0.1/24	—
eth1	—	2000::1/64
eth2	—	2001::1/64

DHCPv4: Базовая настройка

Конфигурация VyOS (шаг 1-2)

```
ankomyagin-gw-01 - PuTTY
[edit]
vyos@vyos# set system host-name ankomyagin-gw-01
[edit]
vyos@vyos# set system domain-name ankomyagin.net
[edit]
vyos@vyos# set system login user ankomyagin authentication plaintext-password a
ankomyagin

Configuration path: system login user ankomyagin [authentication] is not vali
d
Set failed

[edit]
vyos@vyos# set system login user ankomyagin authentication plaintext-password an
komyagin
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ exit
logout

Welcome to VyOS - ankomyagin-gw-01 ttyS0

ankomyagin-gw-01 login: ankomyagin
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
ankomyagin@ankomyagin-gw-01:~$ configure
[edit]
ankomyagin@ankomyagin-gw-01# delete system login user vyos
[edit]
ankomyagin@ankomyagin-gw-01# commit
[edit]
```

Конфигурация интерфейсов

```
ankomyagin-gw-01 - PuTTY
0::/64
[edit]
ankomyagin@ankomyagin-gw-01# set service router-advert interface eth1 other-conf
ig-flag
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless subnet 2000::0/64
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless common-options name-server 2000::1
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless common-options domain-search ankomyagin.net
[edit]
ankomyagin@ankomyagin-gw-01# commit
[edit]
ankomyagin@ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
ankomyagin@ankomyagin-gw-01# run show configuration
interfaces {
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:74:14:d7:00:00
    }
    ethernet eth1 {
        address 2000::1/64
        hw-id 0c:74:14:d7:00:01
    }
    ethernet eth2 {
        address 2001::1/64
        hw-id 0c:74:14:d7:00:02
    }
    loopback lo {
    }
}
service {
    dhcp-server {
        shared-network-name ankomyagin {
            domain-name ankomyagin.net
            name-server 10.0.0.1
            subnet 10.0.0.0/24 {
                default-router 10.0.0.1
            }
        }
    }
}
```

DHCPv4: Конфигурация сервера

```
ankomyagin@ankomyagin-gw-01# set service dhcp-server shared-network-name ankomya
gin domain-name ankomyagin.net
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcp-server sharen-network-name ankomya
gin name-server 10.0.0.1

Configuration path: service dhcp-server [sharen-network-name] is not valid
Set failed

[edit]
ankomyagin@ankomyagin-gw-01# set service dhcp-server shared-network-name ankomya
gin name-server 10.0.0.1
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcp-server shared-network-name ankomya
gin subnet 10.0.0.0/24 default-router 10.0.0.1
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcp-server shared-network-name ankomya
gin subnet 10.0.0.0/24 range hosts start 10.0.0.2
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcp-server shared-network-name ankomya
gin subnet 10.0.0.0/24 range hosts stop 10.0.0.253
[edit]
ankomyagin@ankomyagin-gw-01# commit
```

Рис. 4: Создание DHCPv4-пула ankomyagin с параметрами

```
ankomyagin@ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
ankomyagin@ankomyagin-gw-01# exit
exit
ankomyagin@ankomyagin-gw-01:~$ show dhcp server statistics
Pool          Size      Leases    Available  Usage
-----
ankomyagin    252        0         252      0%
ankomyagin@ankomyagin-gw-01:~$ show dhcp server leases
IP address    Hardware address  State   Lease start   Lease expiration  Re
maining      Pool      Hostname
-----
ankomyagin@ankomyagin-gw-01:~$
```

Рис. 5: Проверка конфигурации и статистики DHCP

DHCPv4: Работа клиента

Клиент PC1 получает адрес

PC1-ankomyagin - PuTTY

```
Welcome to Virtual PC Simulator, version 0.8.3
Dedicated to Daling.
Build time: Sep  9 2023 11:15:00
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '7' to get help.

Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier - Hardware Type-Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier - Hardware Type-Ethernet MAC Address = 00:50:79:66:68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = ankomyagin.net
```

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```
PC1-ankomyagin - PuTTY
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Ack
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = ankomyagin.net

IP 10.0.0.2/24 GW 10.0.0.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.0.2/24
GATEWAY    : 10.0.0.1
DNS        : 10.0.0.1
DHCP SERVER : 10.0.0.1
DHCP LEASE  : 86264, 86400/43200/75600
DOMAIN NAME : ankomyagin.net
MAC        : 00:50:79:66:68:00
LPORT      : 10004
RHOST:PORT  : 127.0.0.1:10005
MTU        : 1500

VPCS> ping 10.0.0.1 -c 2

84 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=1.531 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=3.624 ms

VPCS>
```

Рис. 7: Появление записи об аренде и тестирование связности

DHCPv6: Stateless (PC2)

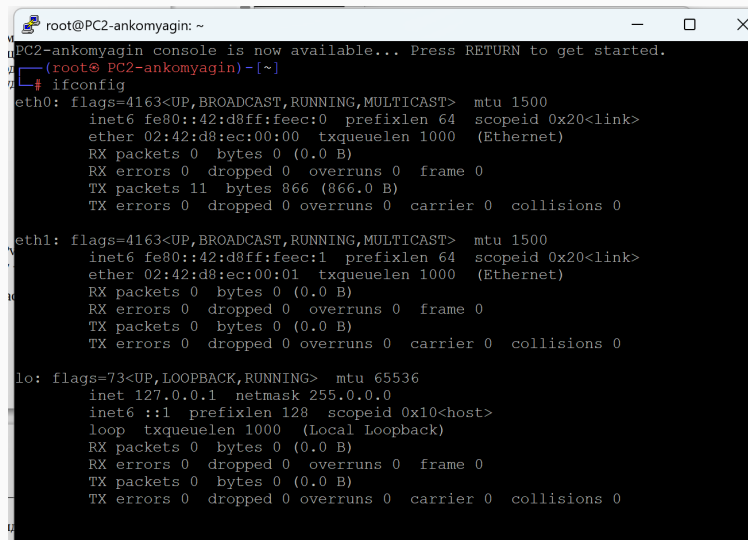
Конфигурация RA и DHCPv6 (stateless)

```
ankomyagin-gw-01 - PuTTY
0::/64
[edit]
ankomyagin@ankomyagin-gw-01# set service router-advert interface eth1 other-conf
ig-flag
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless subnet 2000::0/64
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless common-options name-server 2000::1
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateless common-options domain-search ankomyagin.net
[edit]
ankomyagin@ankomyagin-gw-01# commit
[edit]
ankomyagin@ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
ankomyagin@ankomyagin-gw-01# run show configuration
interfaces {
    ethernet eth0 {
        address 10.0.0.1/24
        hw-id 0c:74:14:d7:00:00
    }
    ethernet eth1 {
        address 2000::1/64
        hw-id 0c:74:14:d7:00:01
    }
    ethernet eth2 {
        address 2001::1/64
        hw-id 0c:74:14:d7:00:02
    }
    loopback lo {
    }
}
service {
    dhcp-server {
        shared-network-name ankomyagin {
            domain-name ankomyagin.net
            name-server 10.0.0.1
            subnet 10.0.0.0/24 {
                default-router 10.0.0.1
            }
        }
    }
}
```

 ankomyagin-gw-01 - PuTTY

```
}  
ethernet eth2 {  
    address 2001::1/64  
    hw-id 0c:74:14:d7:00:02  
}  
loopback lo {  
}  
}  
service {  
    dhcp-server {  
        shared-network-name ankomyagin {  
            domain-name ankomyagin.net  
            name-server 10.0.0.1  
            subnet 10.0.0.0/24 {  
                default-router 10.0.0.1  
                range hosts {  
                    start 10.0.0.2  
                    stop 10.0.0.253  
                }  
            }  
        }  
    }  
}  
    dhcpv6-server {  
        shared-network-name ankomyagin-stateless {  
            common-options {  
                domain-search ankomyagin.net  
                name-server 2000::1  
            }  
            subnet 2000::0/64 {  
            }  
        }  
    }  
}
```

DHCPv6: SLAAC и параметры



```
root@PC2-ankomyagin: ~
PC2-ankomyagin console is now available... Press RETURN to get started.
(root@PC2-ankomyagin)~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:d8ff:feec:0 prefixlen 64 scopeid 0x20<link>
    ether 02:42:d8:ec:00:00 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 11 bytes 866 (866.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:d8ff:feec:1 prefixlen 64 scopeid 0x20<link>
    ether 02:42:d8:ec:00:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рис. 10: ifconfig: link-local адреса и отсутствие глобального адреса

Маршрутизация и первый ping

```
root@PC2-ankomyagin: ~  
  
(root@ PC2-ankomyagin)-[~]  
# route -n -A inet6  
Kernel IPv6 routing table  
Destination                Next Hop                    Flag Met Ref  Use If  
2000::/64                   ::                          UAe  256 1      0 eth0  
fe80::/64                   ::                          U    256 1      0 eth0  
fe80::/64                   ::                          U    256 1      0 eth1  
::/0                        fe80::e74:14ff:fed7:1      UGDAe 1024 1      0 et  
h0  
::1/128                     ::                          Un   0  3      0 lo  
2000::42:d8ff:feec:0/128    ::                          Un   0  2      0 eth0  
fe80::42:d8ff:feec:0/128    ::                          Un   0  3      0 eth0  
fe80::42:d8ff:feec:1/128    ::                          Un   0  2      0 eth1  
ff00::/8                   ::                          U    256 3      0 eth0  
ff00::/8                   ::                          U    256 1      0 eth1  
::/0                        ::                          !n   -1  1      0 lo  
  
(root@ PC2-ankomyagin)-[~]  
# ping 2000::1 -c 2  
PING 2000::1 (2000::1) 56 data bytes  
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=10.8 ms  
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=2.27 ms  
  
--- 2000::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1003ms  
rtt min/avg/max/mdev = 2.268/6.555/10.842/4.287 ms
```

Рис. 11: route -n -A inet6 и ping 2000::1

DHCPv6: Получение параметров

Запуск DHCPv6-клиента (stateless)

```
root@PC2-ankomyagin: ~  
(root@ PC2-ankomyagin)-[~]  
# cat /etc/resolv.conf  
  
(root@ PC2-ankomyagin)-[~]  
# dhclient -6 -S -v eth0  
Internet Systems Consortium DHCP Client 4.4.3-P1  
Copyright 2004-2022 Internet Systems Consortium.  
All rights reserved.  
For info, please visit https://www.isc.org/software/dhcp/  
  
Listening on Socket/eth0  
Sending on Socket/eth0  
Created duid "\000\003\000\001\002B\330\354\000\000".  
PRC: Requesting information (INIT).  
XMT: Forming Info-Request, 0 ms elapsed.  
XMT: Info-Request on eth0, interval 930ms.  
RCV: Reply message on eth0 from fe80::e74:14ff:fed7:1.  
PRC: Done.  
  
(root@ PC2-ankomyagin)-[~]  
# ping 2000::1 -c2  
PING 2000::1 (2000::1) 56 data bytes  
64 bytes from 2000::1: icmp_seq=1 ttl=64 time=3.60 ms  
64 bytes from 2000::1: icmp_seq=2 ttl=64 time=1.78 ms  
  
--- 2000::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1001ms  
rtt min/avg/max/mdev = 1.784/2.694/3.604/0.910 ms  
  
(root@ PC2-ankomyagin)-[~]  
# cat /etc/resolv.conf  
search ankomyagin.net.  
nameserver 2000::1
```

Захват трафика ICMPv6 и NDP

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::e74:14ff:fed7...	ff02::1	ICMPv6	118	Router Advertisement from 0c:74:14:d7:00:01
2	11.473148	2000::42:d8ff:feec:0	2000::1	ICMPv6...	118	Echo (ping) request id=0x0003, seq=1, hop limit=64 (reply in 3) (Send...
3	11.476350	2000::1	2000::42:d8ff:feec:0	ICMPv6...	118	Echo (ping) reply id=0x0003, seq=1, hop limit=64 (request in 2) (Send...
4	12.477177	2000::42:d8ff:feec:0	2000::1	ICMPv6...	118	Echo (ping) request id=0x0003, seq=2, hop limit=64 (reply in 5) (Send...
5	12.478825	2000::1	2000::42:d8ff:feec:0	ICMPv6...	118	Echo (ping) reply id=0x0003, seq=2, hop limit=64 (request in 4) (Send...
6	16.879905	fe80::42:d8ff:feec:0	2000::1	ICMPv6	86	Neighbor Solicitation for 2000::1 from 02:42:d8:ec:00:00
7	16.881463	2000::1	fe80::42:d8ff:feec:0	ICMPv6	78	Neighbor Advertisement 2000::1 (rtr, sol)
8	16.898199	fe80::e74:14ff:fed7...	2000::42:d8ff:feec:0	ICMPv6	86	Neighbor Solicitation for 2000::42:d8ff:feec:0 from 0c:74:14:d7:00:01
9	16.898301	2000::42:d8ff:feec:0	fe80::e74:14ff:fed7...	ICMPv6	78	Neighbor Advertisement 2000::42:d8ff:feec:0 (sol)
10	21.999855	fe80::42:d8ff:feec:0	fe80::e74:14ff:fed7...	ICMPv6	86	Neighbor Solicitation for fe80::e74:14ff:fed7:1 from 02:42:d8:ec:00:00
11	22.001665	fe80::e74:14ff:fed7...	fe80::42:d8ff:feec:0	ICMPv6	78	Neighbor Advertisement fe80::e74:14ff:fed7:1 (rtr, sol)
12	22.023854	fe80::e74:14ff:fed7...	fe80::42:d8ff:feec:0	ICMPv6	86	Neighbor Solicitation for fe80::42:d8ff:feec:0 from 0c:74:14:d7:00:01
13	22.024020	fe80::42:d8ff:feec:0	fe80::e74:14ff:fed7...	ICMPv6	78	Neighbor Advertisement fe80::42:d8ff:feec:0 (sol)

Рис. 13: Wireshark: RA, Echo Request/Reply, Neighbor Solicitation/Advertisement

DHCPv6: Stateful (PC3)

Конфигурация RA (managed-flag) и DHCPv6

```
ankomyagin-gw-01 - PuTTY
}
[edit]
ankomyagin@ankomyagin-gw-01# set service router-advert interface eth2 managed-fl
ag
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateful
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateful subnet 2001::0/64
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateful subnet 2001::0/64 name-server 2001::1
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateful subnet 2001::0/64 2001::0/64 domain-search ankomyagin.net

Configuration path: service dhcpv6-server shared-network-name ankomyagin-stateful subnet 2001::0/64 [2001::0/64] is not valid
Set failed

[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateful subnet 2001::0/64 domain-search ankomyagin.net
[edit]
ankomyagin@ankomyagin-gw-01# set service dhcpv6-server shared-network-name ankomyagin-stateful subnet 2001::0/64 address-range start 2001::100 stop 2001::199
[edit]
ankomyagin@ankomyagin-gw-01# commit
[edit]
ankomyagin@ankomyagin-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
ankomyagin@ankomyagin-gw-01#
```

Интерфейсы PC3 до DHCPv6

```
root@PC3-ankomyagin: ~  
PC3-ankomyagin console is now available... Press RETURN to get started.  
(root@ PC3-ankomyagin)-[~]  
# ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::42:66ff:fe59:a400 prefixlen 64 scopeid 0x20<link>  
    ether 02:42:66:59:a4:00 txqueuelen 1000 (Ethernet)  
    RX packets 13 bytes 1470 (1.4 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 14 bytes 1076 (1.0 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet6 fe80::42:66ff:fe59:a401 prefixlen 64 scopeid 0x20<link>  
    ether 02:42:66:59:a4:01 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(root@ PC3-ankomyagin)-[~]  
#
```

DHCPv6:

Solicit-Advertise-Request-Reply

```
(root@ PC3-ankomyagin) - [~]
# route -n -A inet6
Kernel IPv6 routing table
Destination                Next Hop                    Flag Met Ref  Use If
fe80::/64                   ::                          U    256 1      0 eth0
fe80::/64                   ::                          U    256 1      0 eth1
:::/0                       fe80::e74:14ff:fed7:2      UGDAe 1024 1      0 et
h0
::1/128                     ::                          Un    0  2      0 lo
fe80::42:66ff:fe59:a400/128 ::                          Un    0  3      0 eth0
fe80::42:66ff:fe59:a401/128 ::                          Un    0  2      0 eth1
ff00::/8                   ::                          U    256 4      0 eth0
ff00::/8                   ::                          U    256 1      0 eth1
:::/0                      ::                          !n   -1  1      0 lo

(root@ PC3-ankomyagin) - [~]
# cat /etc/resolv.conf
```

Рис. 16: route -n -A inet6 на PC3: только link-local и маршрут по умолчанию

Работа dhclient (stateful)

```
(root@ PC3-ankomyagin)-[~]
# dhclient -6 -v eth0
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on Socket/eth0
Sending on Socket/eth0
Created duid "\000\001\000\0010\301\345\002BfY\244\000".
PRC: Soliciting for leases (INIT).
XMT: Forming Solicit, 0 ms elapsed.
XMT: X-- IA_NA 66:59:a4:00
XMT: | X-- Request renew in +3600
XMT: | X-- Request rebind in +5400
XMT: Solicit on eth0, interval 1090ms.
RCV: Advertise message on eth0 from fe80::e74:14ff:fed7:2.
RCV: X-- IA_NA 66:59:a4:00
RCV: | X-- starts 1764663653
RCV: | X-- t1 - renew +0
RCV: | X-- t2 - rebind +0
RCV: | X-- [Options]
RCV: | | X-- IAADDR 2001::199
RCV: | | | X-- Preferred lifetime 27000.
RCV: | | | X-- Max lifetime 43200.
RCV: X-- Server ID: 00:01:00:01:30:b1:c1:31:0c:74:14:d7:00:01
RCV: Advertisement recorded.
PRC: Selecting best advertised lease.
PRC: Considering best lease.
PRC: X-- Initial candidate 00:01:00:01:30:b1:c1:31:0c:74:14:d7:00:01 (s: 10105,
p: 0).
XMT: Forming Request, 0 ms elapsed.
XMT: X-- IA_NA 66:59:a4:00
XMT: | X-- Requested renew +3600
XMT: | X-- Requested rebind +5400
XMT: | | X-- IAADDR 2001::199
XMT: | | | X-- Preferred lifetime +7200
XMT: | | | X-- Max lifetime +7500
XMT: V IA_NA appended.
XMT: Request on eth0, interval 1070ms.
RCV: Reply message on eth0 from fe80::e74:14ff:fed7:2.
RCV: X-- IA_NA 66:59:a4:00
RCV: | X-- starts 1764663654
RCV: | X-- t1 - renew +0
RCV: | X-- t2 - rebind +0
RCV: | X-- [Options]
RCV: | | X-- IAADDR 2001::199
RCV: | | | X-- Preferred lifetime 7200.
RCV: | | | X-- Max lifetime 7500.
RCV: X-- Server ID: 00:01:00:01:30:b1:c1:31:0c:74:14:d7:00:01
PRC: Bound to lease 00:01:00:01:30:b1:c1:31:0c:74:14:d7:00:01.

(root@ PC3-ankomyagin)-[~]
```

DHCPv6: Итоги на PC3

Полученная конфигурация

```
(root@ PC3-ankomyagin) - [~]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 2001::199 prefixlen 128 scopeid 0x0<global>
    inet6 fe80::42:66ff:fe59:a400 prefixlen 64 scopeid 0x20<link>
    ether 02:42:66:59:a4:00 txqueuelen 1000 (Ethernet)
    RX packets 18 bytes 2092 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 21 bytes 1828 (1.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::42:66ff:fe59:a401 prefixlen 64 scopeid 0x20<link>
    ether 02:42:66:59:a4:01 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Рис. 18: ifconfig: глобальный адрес 2001::199/128 и link-local

Маршрутизация и DNS

```
root@PC3-ankomyagin: ~  
(root@ PC3-ankomyagin)~[~]  
# route -n -A inet6  
Kernel IPv6 routing table  
Destination                                Next Hop                                Flag Met Ref  Use I  
f  
2001::199/128                              ::                                     Ue   256 1    0 e  
th0  
fe80::/64                                  ::                                     U    256 1    0 e  
th0  
fe80::/64                                  ::                                     U    256 1    0 e  
th1  
::/0                                       fe80::e74:14ff:fed7:2                UGDAe 1024 1    0  
eth0  
::1/128                                    ::                                     Un    0 3    0 1  
o  
2001::199/128                              ::                                     Un    0 2    0 e  
th0  
fe80::42:66ff:fe59:a400/128                ::                                     Un    0 5    0 e  
th0  
fe80::42:66ff:fe59:a401/128                ::                                     Un    0 2    0 e  
th1  
ff00::/8                                   ::                                     U    256 4    0 e  
th0  
ff00::/8                                   ::                                     U    256 1    0 e  
th1  
::/0                                       !n   -1 1    0 1  
o  
(root@ PC3-ankomyagin)~[~]  
# ping 2001::1 -c 2  
PING 2001::1 (2001::1) 56 data bytes  
64 bytes from 2001::1: icmp_seq=1 ttl=64 time=8.75 ms  
64 bytes from 2001::1: icmp_seq=2 ttl=64 time=2.52 ms  
--- 2001::1 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1003ms  
rtt min/avg/max/mdev = 2.522/5.633/8.745/3.111 ms  
(root@ PC3-ankomyagin)~[~]  
# cat /etc/resolv.conf  
search ankomyagin.net.  
nameserver 2001::1  
(root@ PC3-ankomyagin)~[~]
```

DHCPv6: Сервер и трафик

```
ankomyagin@ankomyagin-gw-01# run show dhcpv6 server leases
IPv6 address      State      Last communication      Lease expiration      Remaining
Type              Pool                               IAID_DUID
-----
-----
2001::199         active    2025/12/02 08:20:53    2025/12/02 10:25:53    1:59:35
non-temporary    ankomyagin-stateful  00:a4:59:66:00:01:00:01:30:c1:5d:e5:02:42:6
6:59:a4:00
[edit]
ankomyagin@ankomyagin-gw-01#
```

Рис. 20: show dhcpv6 server leases: активная аренда 2001::199

Захват DHCPv6 и ICMPv6

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::e74:14ff:fed7...	ff02::1:6	ICMPv6	110	Multicast Listener Report Message v2
2	0.159693	fe80::e74:14ff:fed7...	ff02::1:6	ICMPv6	110	Multicast Listener Report Message v2
3	9.928001	fe80::e74:14ff:fed7...	ff02::1	ICMPv6	86	Router Advertisement from 0c:74:14:d7:00:02
4	25.962222	fe80::e74:14ff:fed7...	ff02::1	ICMPv6	86	Router Advertisement from 0c:74:14:d7:00:02
5	41.982195	fe80::e74:14ff:fed7...	ff02::1	ICMPv6	86	Router Advertisement from 0c:74:14:d7:00:02
6	216.634203	fe80::42:66ff:fe59:...	ff02::1:2	DHCPv6	118	Solicit XID: 0x6d11bb CID: 0001000130c15de502426659a400
7	216.688437	fe80::e74:14ff:fed7...	fe80::42:66ff:fe59:...	DHCPv6	186	Advertise XID: 0x6d11bb IAA: 2001::199 CID: 0001000130c15de502426659a...
8	217.726048	fe80::42:66ff:fe59:...	ff02::1:2	DHCPv6	164	Request XID: 0x172108 CID: 0001000130c15de502426659a400 IAA: 2001::19...
9	217.761694	fe80::e74:14ff:fed7...	fe80::42:66ff:fe59:...	DHCPv6	186	Reply XID: 0x172108 IAA: 2001::199 CID: 0001000130c15de502426659a400
10	217.779068	fe80::42:66ff:fe59:...	ff02::1:6	ICMPv6	110	Multicast Listener Report Message v2
11	218.409396	::	ff02::1:ff00:199	ICMPv6	86	Neighbor Solicitation for 2001::199
12	218.663536	fe80::42:66ff:fe59:...	ff02::1:6	ICMPv6	110	Multicast Listener Report Message v2
13	222.009000	fe80::e74:14ff:fed7...	fe80::42:66ff:fe59:...	ICMPv6	86	Neighbor Solicitation for fe80::42:66ff:fe59:a400 from 0c:74:14:d7:00...
14	222.009106	fe80::42:66ff:fe59:...	fe80::e74:14ff:fed7...	ICMPv6	78	Neighbor Advertisement fe80::42:66ff:fe59:a400 (sol)
15	227.304594	fe80::42:66ff:fe59:...	fe80::e74:14ff:fed7...	ICMPv6	86	Neighbor Solicitation for fe80::e74:14ff:fed7:2 from 02:42:66:59:a4:00
16	227.306338	fe80::e74:14ff:fed7...	fe80::42:66ff:fe59:...	ICMPv6	78	Neighbor Advertisement fe80::e74:14ff:fed7:2 (rtr, sol)
17	257.317017	fe80::e74:14ff:fed7...	ff02::1	ICMPv6	86	Router Advertisement from 0c:74:14:d7:00:02
18	490.092452	2001::199	2001::1	ICMPv6...	118	Echo (ping) request id=0x0004, seq=1, hop limit=64 (reply in 21) (Sen...
19	490.097871	2001::1	ff02::1:ff00:199	ICMPv6	86	Neighbor Solicitation for 2001::199 from 0c:74:14:d7:00:02
20	490.098388	2001::199	2001::1	ICMPv6	86	Neighbor Advertisement 2001::199 (sol, ovr) is at 02:42:66:59:a4:00
21	490.100684	2001::1	2001::199	ICMPv6...	118	Echo (ping) reply id=0x0004, seq=1, hop limit=64 (request in 18) (Sen...
22	491.096043	2001::199	2001::1	ICMPv6...	118	Echo (ping) request id=0x0004, seq=2, hop limit=64 (reply in 23) (Sen...
23	491.097395	2001::1	2001::199	ICMPv6...	118	Echo (ping) reply id=0x0004, seq=2, hop limit=64 (request in 22) (Sen...
24	495.591501	fe80::42:66ff:fe59:...	2001::1	ICMPv6	86	Neighbor Solicitation for 2001::1 from 02:42:66:59:a4:00
25	495.591550	fe80::42:66ff:fe59:...	fe80::e74:14ff:fed7...	ICMPv6	86	Neighbor Solicitation for fe80::e74:14ff:fed7:2 from 02:42:66:59:a4:00
26	495.593087	2001::1	fe80::42:66ff:fe59:...	ICMPv6	78	Neighbor Advertisement 2001::1 (rtr, sol)
27	495.593379	fe80::e74:14ff:fed7...	fe80::42:66ff:fe59:...	ICMPv6	78	Neighbor Advertisement fe80::e74:14ff:fed7:2 (rtr, sol)
28	500.622687	fe80::e74:14ff:fed7...	fe80::42:66ff:fe59:...	ICMPv6	86	Neighbor Solicitation for fe80::42:66ff:fe59:a400 from 0c:74:14:d7:00...
29	500.623286	fe80::42:66ff:fe59:...	fe80::e74:14ff:fed7...	ICMPv6	78	Neighbor Advertisement fe80::42:66ff:fe59:a400 (sol)

Рис. 21: Wireshark: Solicit, Advertise, Request, Reply, Echo ping, NDP

Результаты

- DHCPv4: автоматическая выдача адреса PC1;
- Stateless-DHCPv6: SLAAC + параметры DNS для PC2;
- Stateful-DHCPv6: полная выдача адреса 2001::199 для PC3;
- Анализ трафика ICMPv6, NDP и DHCPv6.

- DHCPv4 обеспечивает централизованную выдачу параметров IPv4;
- SLAAC позволяет хостам независимо генерировать адреса в IPv6;
- DHCPv6 дополняет SLAAC, выступая в роли поставщика параметров или полного адреса;
- Протоколы NDP и DHCPv6 обеспечивают корректную работу IPv6-сетей.